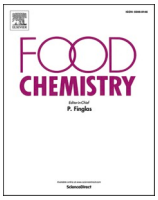
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Spectral reconstruction from RGB image to hyperspectral image: Take the detection of glutamic acid index in beef as an example

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ABSTRACT

The use of spectral reconstruction (SR) to recovery RGB images to full-scene hyperspectral image (HSI) is an important measure to achieve real-time and low-cost HSI applications. Taking the detection of glutamic acid index for 360 beef samples as an example, the feasibility of using 11 state-of-the-art reconstruction algorithms to achieve RGB to HSI in complex food systems was investigated. The multivariate correlation analysis was used to prove that RGB is a projection of three-channel comprehensive coverage wide-band information. The comprehensive quality attributes (PSNR-Params-FLOPS) was proposed to determine the optimal reconstruction model (MST++, MST, MIRNet, and MPRNet). Moreover, SSIM values and t-SNE were introduced to evaluate the consistency of the reconstruction results. Finally, Lightweight Transformer was used to establish the detection models of Raw-HSI, RGB and SR-HSI for the prediction of glutamic acid index for beef. The results showed that the MST++ model exhibited the best performance in SR, with RMSE, PSNR, and SSIM values of 0.015, 36.70, and 0.9253, respectively. Meanwhile, the prediction effect of MST++ ($R_P^2 = 0.8422$ and RPD = 2.46) reconstructed was close to the Raw-HSI ($R_P^2 = 0.8526$ and RPD = 2.69). The results provide practical application scenarios and detailed analysis ideas for RGB-to-HSI.

1. Introduction

The nutrition and flavor attributes of meat are not only an important reference to consumer preferences, but also the key factor driving purchase decisions for consumer (Dong, Bi, Hao, & Wang, 2024; Wang et al., 2020). The flavor of high-quality meat mainly from the unique compounds produced by enzyme reactions, including water-soluble compounds, peptides, fatty acids, and free amino acids (FAAs) (Bi et al., 2022; Dong et al., 2023). Usually, nutrition and flavor parameters are needed to be considered and evaluated during the processing and production of meat (Wan et al., 2023; Zhang et al., 2022). In particular, as the most important umami FAAs, glutamic acid (Glu) can alleviate food acidity and alkalinity by enhancing the flavor of beef (Yin et al., 2022). Yin et al. (2022) showed that Glu mixed with a small amount of flavor nucleotides could improve the flavor of food, making it more rounded and intense. Previous research has shown that meat nutritional and flavor parameters are associated with overall quality and acceptability (Dong et al., 2023; Zhang, Ma, et al., 2022). Therefore, it is necessary to analyze meat nutritional and flavor parameters during processing and storage. Traditional detection methods include automatic amino acid analyzer, high performance liquid chromatography and gas chromatography (Dong et al., 2022; Dong et al., 2023; Wang et al., 2020; Wang, Wang, Dong, & Wang, 2021). Although their detection accuracy is high, there are certain hazards and pollution (Liu et al., 2023; Lv et al., 2023). Therefore, it is urgent to develop a rapid, non-destructiveand low-cost quantitative detection method for meat nutritional and flavor parameters.

In recent years, emerging technologies including hyperspectral imaging (HSI), fluorescence microscopic HSI, nuclear magnetic resonanceand Raman spectroscopyhave been widely used in the detection of meat quality (Hao et al., 2023; Wan et al., 2023; Wang et al., 2020). Among, as a general equipment for analyzing and recognizing scene information, HSI has a wide range of application prospects (Dong et al., 2022; Wan et al., 2023; Wang et al., 2021). By capturing objects rich

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spatial and spectral information, HSI has become an important means of qualitative and quantitative analysis of substances (Lv et al., 2023; Zhang, Ma, et al., 2022). HSI has achieved important research results in the fields of remote sensing science (Liu, Bruning, Garnett, & Berger, 2020), agricultural management (Zhao et al., 2020), biomedicine (Calin, Parasca, Savastru, & Manea, 2014; Sun et al., 2022), environmental monitoring (Hu, Hou, Ming, Meifang, & Chen, 2023) and food safety (Liu et al., 2023). However, the complexity and high cost of acquiring HSI data hinder its application and promotion in consumer electronics products (Cai et al., 2023; Shi, Chen, Xiong, Liu, & Wu, 2018; Zhang et al., 2022). For example, its application in daily food safety inspections (Liu et al., 2023), intelligent pollution monitoring (Fang, Luo, & Naidu, 2023), agricultural product quality control (Wang et al., 2023) and point-of-care medical screening (Zhao et al., 2020) has been hindered. For this purpose, some scholars have conducted research on obtaining high-quality HSI at low cost in complex scenes (Alvarez-Gila, van de Weijer, & Garrote, 2017; Cai et al., 2022; Cai et al., 2023; Shi et al., 2018; Shukla, Upadhyay, Sharma, Chinnusamy, & Kumar, 2022; Zou & Wei, 2021). Considering that smart cameras are ubiquitous in daily life worldwide, and it can easily capture many high-quality color images (Cai et al., 2022; Zamir et al., 2021; Zhang, Su, et al., 2022). Therefore, the spectral reconstruction (SR) strategy of using RGB images to recovery HSI provides the best solution. It converts complex HSI image acquisition into real-time and low-cost reconstruction calculations.

SR algorithm is an ill-posed inverse problem since a large amount of spectral information is lost when RGB sensors capture the light (Alvarez-Gila et al., 2017; Arad et al., 2020; Cai, Lin, Lin, et al., 2022; Zhang, Su, et al., 2022). To solve this problem, many efficient algorithms provide solutions (Arad et al., 2018; Arad et al., 2020; Arad et al., 2022; Cai et al., 2023; Cai, Lin, Hu, et al., 2022; Fu et al., 2022; Hu et al., 2023; Zamir et al., 2021; Zou & Wei, 2021). Generally, the existing SR methods can be divided into two categories: prior-based method and data-driven method (Arad et al., 2018; Zhang, Su, et al., 2022). Both two methods take advantage of the redundancy of image data but differ in fundamentals. (1) The prior-based method establishes relatively shallow learning models or sparse coding from HSI statistical information (e.g., sparsity, spatial structure similarity, and spectral correlation) (Akhtar & Mian, 2018; Arad et al., 2018; Cai, Lin, Lin, et al., 2022). It seeks reasonable solutions in a more constrained space (Hu et al., 2023). Nonetheless, the priori-based method is almost hand-crafted and relies on a known spectral response function. It leads to the SR method of expression ability being poorer, and the performance is limited (Zhang, Su, et al., 2022). (2) Data-driven methods learn end-to-end mapping relationships between HSI and RGB images by training on large data (Shukla et al., 2022), which use the advantages of abstract feature extraction to identify the hidden layer mapping relationships and find more accurate solutions (Fu et al., 2022). It is worth noting that simulated reconstructions using data-driven methods on numerous open-source HSI datasets indicate that there is highly complex mapping correlations between pixel values in HSI and RGB images (Cai, Lin, Hu, et al., 2022; Cai, Lin, Lin, et al., 2022; Shukla et al., 2022). This provides algorithmic support for SR of RGB-to-HSI.

In previous investigations, SR has found limited application in specific agricultural and crop products. For instance, Fu et al. (2022) conducted a study on SR of maize diseases based on RGB images combined with HSCNN+ algorithm. Experimental results demonstrated that the reconstructed HSI efficiently improved detection accuracy compared with raw RGB image in tested scenarios. Especially in complex environment scenario, the detection accuracy was increases by 6.14 %. Similarly, Zhao et al. (2020) effectively realizes the prediction of tomato quality parameters ($R^2 > 0.5$) by improving the HSCNN-R model. Furthermore, Hu et al. (2023) proposed the reconstruction of multi-scale atrous residual convolution network, which effectively solved the problem of high-cost Marine water quality monitoring. However, these studies represent only a fraction of the potential applications in the agricultural and food domain, indicating that SR has a significant opportunity for further exploration and innovation.

Therefore, the detection of Glu index of FAAs in beef was taken as an example in this study. The feasibility of state-of-the-art (SOTA) spectral reconstruction algorithm for quantitatively reconstructing of complex food system images was discussesed in this paper. The specific objectives are as follows: (I) sample data set preparation: HSI and RGB image data and chemical values of 360 beef samples were collected; (II) Multivariate data analysis: spectral data correction and dimensionality reduction were realized by utilizing spectral reflectivity index visualization, spectral preprocessing, characteristic wavelength extraction and other technologies. (III) Multivariate correlation analysis: the projection of RGB hidden space wide-band information onto multiple narrow-band to form HSI was discussed by using multivariate correlation scatter matrix. (IV) Spectral reconstruction: 11 SOTA algorithms combined with comprehensive quality attributes and consistency assessment were used to determine the optimal reconstruction model for beef application scenarios. (IV) quantitative model: the Lightweight Transformer algorithm was proposed to evaluate the predictive performance of the Raw-HSI, RGB, and SR-HSI. Overall, the research will provide a more portable and efficient solution for the quantitative analysis of complex food systems. The overall research approach is shown in Scheme 1.

2. Materials and methods

2.1. Sample preparation and chemical value determination

2.1.1. Sample preparation

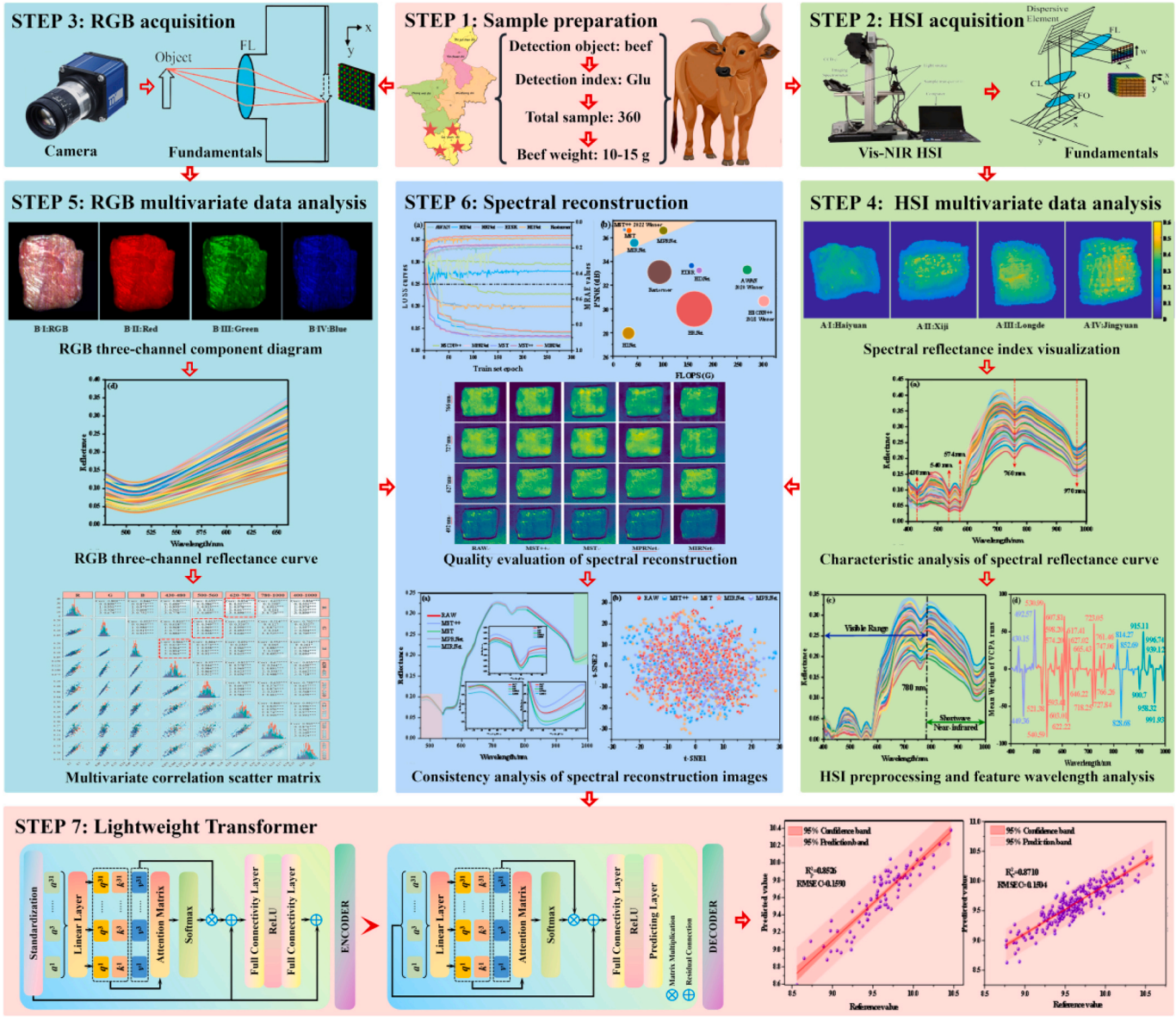
Fresh beef samples were collected from three parts (foreleg (FL), hindleg (HL) and longissimus lumborum (LL)) of four producing regions (Jingyuan (JY), Haiyuan (HY), Longde (LD) and Xiji (XJ)) in Ningxia Hui Autonomous Region, China. After vacuum packing, the samples were put into a portable refrigerated incubator and transported to the Meat Processing and Quality Safety Control Laboratory of Ningxia University. Firstly, the fat and connective tissue present on the surface of the meat were removed. Then, the beef was cut into 40 × 40 × 20 mm (length × width × thickness) pieces, and ensure that the weight of each piece of meat was maintained at 10–15 g. Finally, 360 samples were obtained and placed in sealed bags. It is stored in a refrigerator at −80 °C to prevent potential degradation or changes in the sample. According to the experiment progress, the spectral and chemical values of samples were measured simultaneously.

2.1.2. Determination of chemical value

The method for detecting FAAs in samples of Bi et al. (2022) was improved. Take 2.00 g well-mixed sample and add 0.02 mol/L hydrochloric acid for dissolution and constant volume. The C18 column was activated by adding 5.00 mL methanol and water, respectively. After filtration, 2.5 mL sample solution was absorbed, and 0.02 mol/L hydrochloric acid was added to 5.00 mL. The sample to be measured was processed by a filter membrane with a pore size of 0.45 μm. The specific parameters of automatic amino acid analyzer are set as follows: the wavelength used is 570 and 440 nm; the injection volume was 20.00 μL; the reaction temperature reached 135 ± 5 °C; the separation column was 57 °C; and the chromatographic column was a sulfonate-type cationic resin separation column. By comparing the retention time and peak area of each amino acid standard, the quantitative analysis of each amino acid was carried out according to Eq. 1.

$$X = \frac{C \times V \times N}{M} \tag{1}$$

where X is the content of FAAs in the sample, mg/100 g; M is the sample weighing amount; N is the sample dilution factor; V is the Volumetric volume after hydrolysis; C is the concentration of each FAA calculated on the instrument.



Scheme 1. Flowchart of the main steps in the data analyses and model establishment.

2.2. Data set acquisition system

The Vis-NIR system manufactured by Zolix Instruments Co., Ltd. acquired the HSI data set. This equipment can obtain both HSI and RGB images, which effectively avoids pixel position deviation. The provided data has a spatial resolution of 2.8 nm and a pixel size of 1004 × 301, among which the HSI data has 125 spectral bands from 400 to 1000 nm. The acquisition system mainly composes an imaging spectrometer, CCD camera, halogen lamp, electronic displacement platform, and computer. In order to ensure that the obtained image did not lose frame, the system parameters were determined as follows after multiple experiments: the exposure time of the CCD camera was 10 ms, the electronic displacement speed was 16 mm/s, the object distance was 385 mm, the light source steady current was set to 6.0 A, and the scanning length was 70 mm. Black and white correction of HSI data is as follows:

$$R = \frac{O - B}{W - B} \times 100\% \tag{2}$$

where O is the original spectral image; W is the all-white calibration image; B is the all-black calibration image; and R is the calibrated spectral image. The B was obtained by covering the camera lens cap (0% reflectance), and the W was obtained by a white plate made of polytetrafluoroethylene (PTFE) material (>99 % reflectance).

2.3. The fundamentals of spectral reconstruction

Usually, HSI has hundreds of narrow spectral bands, while RGB images have only three wide spectral channels. In principle, because different objects have different reflectivity to electromagnetic waves of different wavelengths, imaging equipment can capture the reflection spectral of the material surface and convert it into an image. According to the Lambertian assumption (Koppal, 2014), the process of forming an RGB image is shown in Eq. 2:

$$I_C(x,y) = \int_{\lambda 1}^{\lambda 2} L(\lambda) \times R(x,y,\lambda) \times E_C(\lambda) \times d\lambda \tag{3}$$

where I_c represents the 3 channels of a RGB image (c = R, G, B); L (λ) is the illumination spectrum; R (x, y, λ) is the spectral reflectance of the object; Ec (λ) is the spectral response function; λ is the wavelength. It should be stressed that the HSI image H (x, y, λ) is defined as the product

of the spectral reflectance of the substance and the illumination spectrum. Therefore, the relationship between HSI and RGB images is shown in Eq. 3:

$$I_C(x,y) = \int_{\lambda 1}^{\lambda 2} H(x,y,\lambda) \times E_C(\lambda) \times d\lambda \tag{4}$$

Discretizing Eq. 3 yields:

$$I_C(x,y) = \sum_{i=0}^{n} H(\lambda_n) \times E_C(\lambda_n) \tag{5}$$

where λ_n represents the sampled bands, in our study, n has a value of 125 and corresponds to the wavelength range of 400–1000 nm. The challenge in SR is to reverse the above forward formation image model in the optimization framework to find the multi-channel data cube H (λ_n).

2.4. Multivariate data analysis

The high dimensionality and multi-channel of HSI data provide a deeper understanding of the image content. However, it also causes high complexity and intense noise in the data. Therefore, the multivariate analysis of spectral data is crucial to reveal the inherent rules of reconstruction and establish a reliable prediction model. The primary analysis methods are as follows:

(I) Spectral reflectance index visualization: the spectral reflectance index visualization was generated based on selecting the best region of interest image and coupling calibrated HSI (Wan et al., 2023). It combines spectral and spatial information to accurately reflect the objects chemical composition and spatial distribution. The chemical composition difference of pixels between HSI and RGB images were ntuitively showed.

(II) Sample set division: in order to avoid over-fitting problems in the process of model establishment, the random sampling (RS) algorithm proposed by Wei, He, Zheng, and Ye (2020) was referenced. In the spectral reconstruction model, HSI and RGB images were divided into the training set and testing set according to 7:3. In the quantitative prediction model, HSI, RGB, and SR-HSI images were divided into the calibration set, cross-validation set, and prediction sets according to 6:2:2.

(III) Data preprocessing: HSI data are affected by environmental interference and instrument noise, which may lead to wavelength shifts in data collected at different times. Therefore, the preprocessing of spectral data can improve the spectral resolution of overlapping data and minimize the noise impact the instrument brings. The preprocessing of Baseline was mainly used to eliminate the baseline drift of HSI data.

(IV) Feature wavelength extraction: the key factors such as variable contribution, data weighting and data compression are comprehensively considered in the extraction of feature wavelengths. The competitive adaptive re-weighted sampling (CARS) was selected to extract the feature variables. The main advantage of CARS is that it can build a correction model for each variable individually at run time and calculate the weight of the regression coefficient in each variable. It avoids the subjective choice adopts the exponential decreasing function to reduce the variable space, further combined with adaptive re-weighting sampling to reduce variables (Yun et al., 2015).

2.5. Spectral reconstruction algorithm

Deep learning methods have excelled in numerous machine vision tasks and have been successfully used for HSI reconstruction. This study compared 11 SOTA reconstruction methods in *NTIRE 2018, 2020, and 2022 Spectral Reconstruction Challenge* (Arad et al., 2018; Arad et al., 2020; Arad et al., 2022). These include three snapshots compressive imaging (SCI) reconstruction methods (MST, MST++, and HDNet), three classical convolution reconstruction algorithms (HSCNN++, AWAN, and HRNet), and five natural image recovery models (MIRNet, MPRNet, Restormer, HINet, and EDSR). It should be emphasized that HSCNN++, AWAN and MST++ were the winners of *NTIRE 2018, 2020, and 2022 Spectral Reconstruction Challenge*, respectively. The codes of all algorithms are implemented based on Cai, Lin, Lin, et al. (2022). Code and pre-trained models are publicly available at *https://github.com/caiyuanhao1998/MST-plus-plus*.

To ensure fair evaluation of the performance of the reconstruction algorithms, 11 reconstruction algorithms were run under the same hardware environment. Parameter settings are optimized according to Cai, Lin, Lin, et al. (2022), specific parameters are as follows: Adam solver was selected for parameter optimization; batch size was 20; the first order attenuation index β_1 was 0.900; the second-order attenuation index β_2 was 0.999; the learning rate was initialized as 0.0004 and the Cosine Annealing scheme was adopted for 300 epochs. When the training loss did not decrease significantly, the training was stopped. The size of all images is 400 × 301 × 31.

All data analysis was performed on a 12th Gen Intel (R) Core (TM) i7–12,700 CPU with a main frequency of 2.9 GHZ, 16 GB of RAM, NVIDIA GeForce RTX 3090, and 24 GB memory. R-Studio was used for the multi-variable correlation scatter matrix, Python 3.10 was used for the reconstruction algorithm, and Origin 2022a was used for all figures.

2.6. Lightweight transformer

With the development of natural language processing and advanced visual tasks, Transformer technology shows great potential (Cai, Lin, Hu, et al., 2022). Transformer uses a self-attention mechanism to process image tasks, which can establish global associations in the whole image and capture long-distance dependencies between different parts of the image. At the same time, the Transformer can provide better interpretability because of the characteristics of the self-attention mechanism, making the model decision more convincing. Compared with the HSI of remote sensing with more information, the beef spectral images collected in this study are relatively simple. Therefore, a Lightweight Transformer regression model was constructed to detection beef amino acid content by adopting an encoder-decoder structure, and a self-attention mechanism and residual connection mechanism were introduced. Encoder-decoder is a common neural network architecture, in which two modules work together to encode the input sequence and decode the output sequence, respectively (Wang, Wang, Wang, & Wang, 2024). In the encoder part, the data was refined firstly through the multi-self attention mechanism and residual connection mechanism. Among, the flexibility of multi-self attention mechanism enables the model to emphasize more information characteristics, and the residual connection degradation mechanism is used to prevent network degradation and reduce the gradient problems. Then, regularization is performed using the normalization layer and dropout to prevent overfitting of the model. Finally, an activation function (ReLU) is placed between the two feed-forward fully connected layers. This structure is critical to the model to learn complex nonlinear relationships. At the end of the encoder, the residual connection adds the original input data and full connection layer data into the decoder. In order to achieve the goal of quantifying Glu content, the second feed-forward fully connected layer in the decoder is modified as the output layer of the regression model. The Lightweight Transformer network framework is shown in Fig. S1.

2.7. Multivariate evaluation indicators

Evaluating model performance is a critical step in solving a particular problem. A well-chosen performance metric not only quantifies the model generalization ability, but is also essential for efficient solution selection. At the same time, different indicators need to be weighed according to the specific problem and application scenario. This study

involves the evaluation of performance indicators for quality evaluation of SR, consistency evaluation of SR, and spectral quantitative model evaluation.

(I) Quality evaluation of SR: when evaluating the quality of SR, the robustness, error and distortion of spectral feature extraction by the reconstruction algorithm should be considered. The mean relative absolute error (MRAE) loss function between the reconstructed and raw HSI was adopted as the objective. The other evaluation metrics were root mean square error (RMSE) and peak signal to noise ratio (PSNR) (Yi, Zhao, Yun, Borjigin, & Dong, 2024). They were defined as follows:

$$MRAE = \frac{\sum_{i,c} \frac{|P_{gtic} - P_{recic}|}{P_{gtic}}}{|P_{gt}|} \tag{6}$$

$$RMSE = \sqrt{\frac{\sum_{r,c} (P_{gtic} - P_{recic})^2}{|P_{gt}|}} \tag{7}$$

$$PSNR = 20log_{10}\left(\frac{1}{RMSE}\right) \tag{8}$$

where P_{gtic} and P_{recic} represent the value of spectral channel c of the i-th pixel in the truth-value image and the reconstructed image, respectively. $|P_{gt}|$ is the size of the truth-value image (number of pixels x number of spectral channels). The smaller the MRAE and RMSE values and the larger the PSNR values, the closer the reconstruction results are to the actual spectra, indicating the higher the quality of spectral reconstruction.

(II) Consistency evaluation of SR: the structural similarity index (SSIM) is a common index used to quantify the structural similarity between images (Cai, Lin, Hu, et al., 2022). SSIM values range from 0 to 1, with higher values representing more similar images. When the SSIM value is 1, the images are identical. The formula for calculating the SSIM value is as follows:

$$SSIM(x,y) = [L(x,y)^{\alpha}]\left[C(x,y)^{\beta}\right][S(x,y)^{\gamma}] \tag{9}$$

where α, β, and γ represent the weight of each indicator; $L(x,y)$, $C(x,y)$, and $S(x,y)$ correspond to luminance, contrast, and structural difference functions, respectively.

In addition, the stability of reconstruction schemes was discussed by using t-distributed stochastic neighbor embedding (t-SNE) (Maaten & Hinton, 2008). t-SNE is widely used to map high-dimensional datasets into two or three-dimensional space, which visually showes the structure and patterns of the data. It can maintain local relationships between data points, so that similar data points remain close to each other in low-dimensional space. It is important for revealing underlying structures and patterns in the data.

(III) Spectral quantitative model evaluation: the evaluation of quantitative model provides scientific basis for decision making. In order to better evaluate the robustness and accuracy of the model, coefficient of determination (R^2), root mean square error (RMSE), and residual prediction deviation (RPD) were selected as evaluation indexes. Generally, a good model should have high R^2 and RPD values, and have low RMSE values. The equation for evaluation indicators as follows:

$$R^2 = 1 - \frac{\sum_{i=1}^{n} (y_i - \hat{y}_i)^2}{\sum_{i=1}^{n} (y_i - \overline{y_i})^2} \tag{10}$$

$$RMSE = \sqrt{\frac{1}{n}\sum_{i=0}^{n} (\hat{y}_i - y_i)^2} \tag{11}$$

$$RPD = \frac{SD}{RMSE} \tag{12}$$

3. Results and discussion

3.1. Glutamic acid content analysis

The statistical analysis results of Glu content in beef samples from different regions (JY, HY, LD, and XJ) and parts (LL, FL, and HL) are shown in Fig. 1. The average contents of JY, HY, LD, and XJ in different producing regions were 4.179 ± 0.386, 3.524 ± 0.482, 3.641 ± 0.267, and 3.558 ± 0.337 mg/100 g, respectively. The variability of Glu data in JY and HY production regions was the highest, with data ranges of 3.616–5.948 and 2.603–5.036 mg/100 g, respectively. The variability in LD and XJ production regions was similar. In different parts, the content of LL (3.906 ± 0.299 mg/100 g) was the highest, the content of FL (3.788 ± 0.434 mg/100 g) and HL (3.65 ± 0.622 mg/100 g) was similar, and the coefficient of variation increased. The reason might be related to the energy provided by FAAs in the process of catabolism. When the supply of FAA exceeds the body needs, they will generte α-ketoic acid, oxaloacetic acid, fumaric acid, acetyl coA, pyruvate and other intermediate metabolites under the combined deamination, then participate in the tricarboxylic acid cycle and the de-novo synthesis of sugars and fatty acids (Dong et al., 2024). The exercise on the LL of beef is smaller than that on the FL and HL, with more fat content and higher glycolytic activity. Therefore, more FAA are needed to provide precursor materials for fatty acid and sugar synthesis. This result is consistent with the research results of Wang et al. (2020) and Dong et al. (2024) on the research for mutton fatty acids and beef amino acids.

3.2. The visualization for HSI and RGB image channel

RGB images are synthesized by projecting the information of a wide-band filter through the hidden space and into three-channels (Zhang, Su, et al., 2022). Then, the broadband information is projected onto multiple narrow bands through a new hidden space to form HSI, which is essentially compressed perception (Hu et al., 2023). Image visualization provides ideas for visually comparing compression perception properties and channels differences (Wan et al., 2023). The visualization of the HSI and RGB image channels is shown in Fig. 2. The reflectance values of HSI are distributed between 0–0.6, and the RGB colors produced by different bands are significantly different. The visualization of HSI

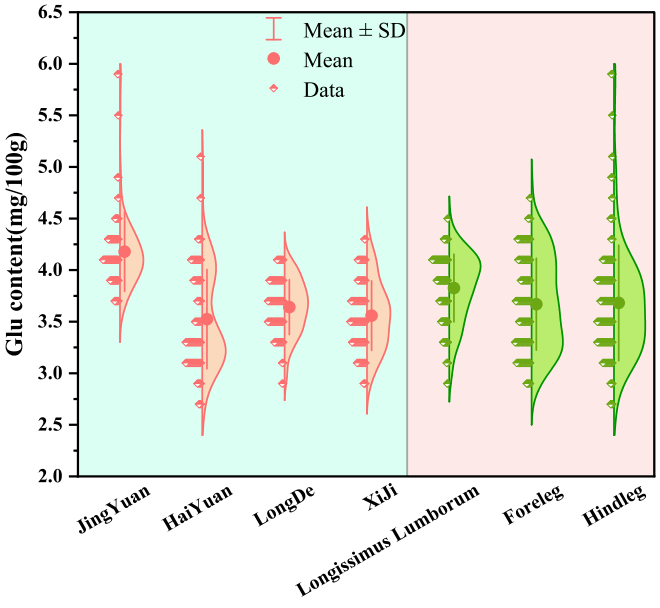


Fig. 1. Statistical analysis of Glu content (mg/100 g) in yellow beef (n = 360) in different parts and regions. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

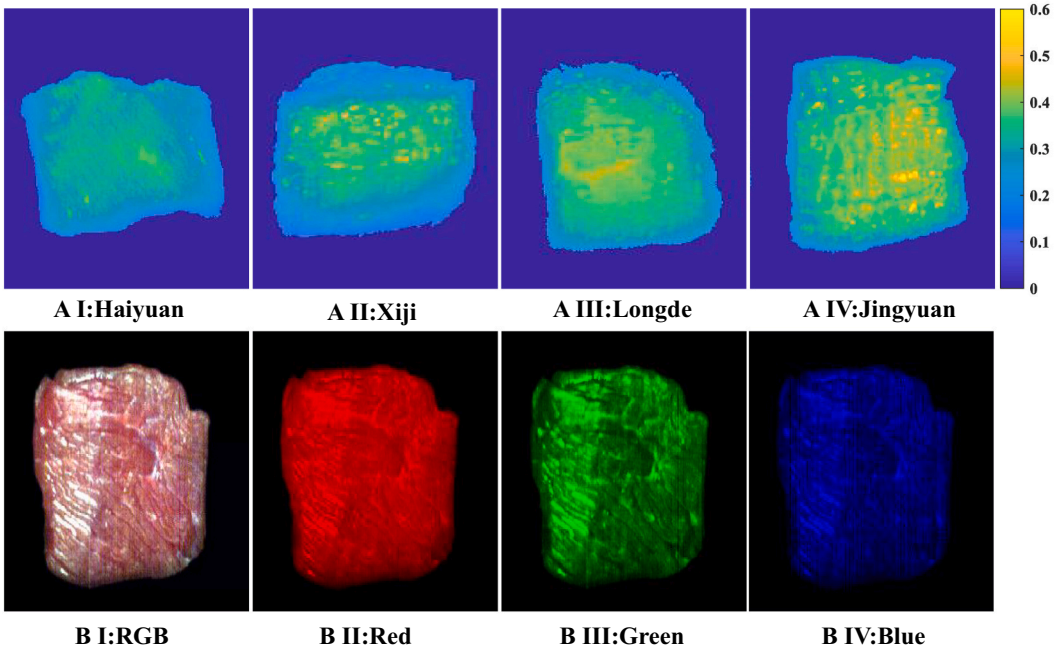


Fig. 2. Visualization of HSI and RGB image channels. (A) Visualization of HSI reflectance index for different producing regionregion; (B) Visualization of RGB three-channel.

reflectance index of beef samples from four production regions is shown in Fig. 2(A). When the color of the beef sample was closer to yellow, the reflectivity index was larger, and vice versa. The beef samples from the Longde and Jingyuan producing regions showed more yellow color, indicating a higher average reflectance. This was not only related to the differences in the chemical composition of the sample pixels, but also closely related to the feeding method and physicochemical properties of muscles (Dong, Bi, et al., 2022).

Fig. 2(B) generates the three-channel RGB component diagram. In the CIE 1931 color space, the green, blueand red components were correspond to 480, 520 and 660 nm point wavelengths, respectively. Comparing the HSI and RGB images, obviously the HSI contained multi-channel information showed detailed spectral details, which could reflect the chemical composition of the object and its spatial distribution. Notably, the RGB component map had unique advantages in visually displaying color information. In summary, the intuitive comparison between HSI and RGB images could reveal the existence of compression perception, but the inherent connection between the two images was not reflected. Therefore, spectral curves needed to be further extracted for analysis.

3.3. Characteristic analysis of spectral reflectance curve

The reflectance values from HSI and RGB images were extracted to compare and confirm their differences. Fig. 3(a) shows the spectral reflectance curve of 360 beef samples. The overall trend of the reflectance curve was similar, but the spectral longitudinal displacement was quite different. Specifically, compared with 600–1000 nm, 400–600 nm had a narrower peak amplitude frequency and a lower reflectivity value. Among, there were major absorption bands at 430, 540, 574, 760, and 970 nm, which were caused by the absorption of energy at specific wavelengths by chemical bonds in beef. The absorption band at 430 nm was the Soret band (B band) arising from an intense transition of $\pi \rightarrow \pi^*$, which was observed in all conjugated porphyrin rings (Andr'es et al., 2008). Absorption bands at 540 and 574 nm were associated with hemoglobin, methemoglobin, and deoxymyoglobin in beef respiratory pigments (Dong et al., 2023). Absorption bands at 760 and 970 nm were attributed to third-overtone and second-overtone of O—H stretching vibrations in water, respectively (Liu et al., 2023). Fig. 3(b) shows the spectral curves of RGB at 480, 520, and 660 nm channels. It can be observed that the overall trend of reflectance of RGB is similar to that of HSI, and the reflectance intensity of corresponding points is the similar. It makes it possible to map the wide-band information in the hidden space onto multiple narrow-bands.

3.4. Characteristic analysis of hyperspectral data

The key factor in the accuracy of RGB-to-HSI reconstruction is to find the end-to-end mapping relationship. To improve data quality and enhance feature information, the Baseline algorithm was selected for spectral preprocessing. Fig. 3(c) shows the spectral image after Baseline preprocessing. It is worth noting that Baseline effectively repairs the noise data at beginning and end of the spectral curves. The process of spectral preprocessing did not change the overall trend and peak position of spectral curves.

The ultimate goal of the research is to recreate the natural RGB image into HSI, but the redundant bands cause time-consuming reconstruction calculations. Selecting the objective feature wavelength can improve the training efficiency and reduce the training cost of the reconstructed image. In this study, the CARS algorithm was used to calculate the average weight value of each wavelength, and the wavelength with a larger weight value was selected as the feature wavelength. Fig. 3(d) shows the average weight distribution diagram of CARS algorithm and the 31 corresponding feature wavelength positions. The selected wavelengths were evenly distributed in the whole band range, including 430.15, 449.36, 492.57, 521.38, 530.99, 540.59, 574.20, 593.41, 598.20, 603.01, 607.81, 617.41, 622.22, 627.02, 646.22, 665.43, 718.25, 723.05, 727.84, 747.06, 761.46, 766.26, 814.27, 828.68, 852.69, 900.7, 915.11, 939.12, 958.32, 991.93 and 996.74 nm. The number of features accounts for 25 % of the total wavelength.

The absorption bands at 430.15, 449.36 and 492.57 nm were related to hemoglobin and methemoglobin in beef respiratory pigment (Dong et al., 2023). Likewise, the spectral features were very similar to Wan et al. (2023), Liu et al. (2023), and Cozzolino and Murray (2004) reported in other meat samples, as the primary pigment of those meat samples from different species were the same. The $\pi \rightarrow \pi^*$ electron

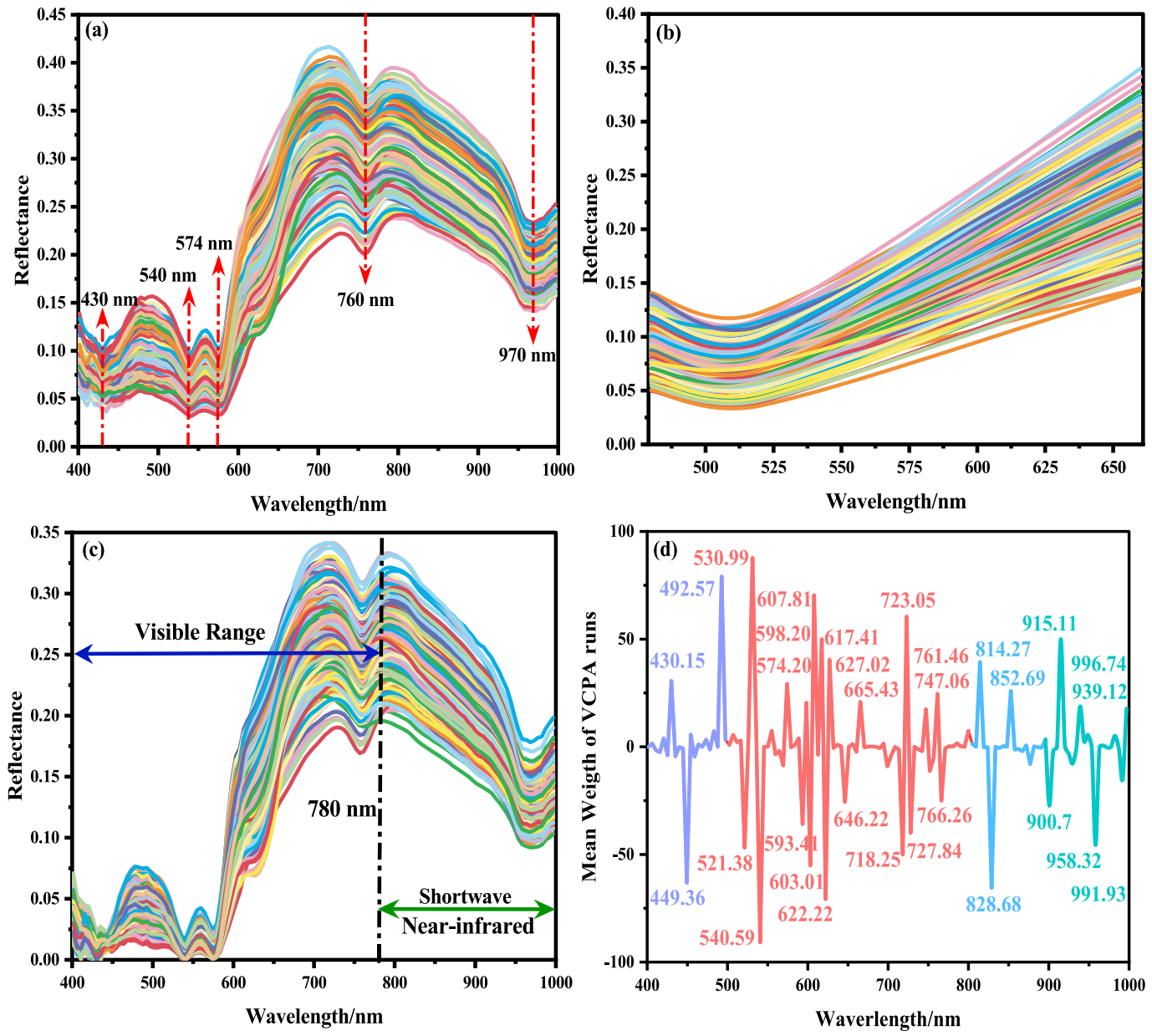


Fig. 3. Curves of HSI and RGB data. (a) reflectance curve of HSI; (b) reflectance curve of RGB image three-channel; (c) HSI data preprocessing; (d) HSI average weight distribution diagram.

transitions occur around 500–700 nm (Q band) are the chromophore and active site of Mb in the heme group (Smulevich et al., 2007). The absorption bands at 700–900 nm mainly were related to the stretching vibrations of O—H in water (Liu et al., 2023). Notably, the absorption peaks of the selected weight values were concentrated between 500 and 800 nm (19 bands) and 900–1000 nm (6 bands). Furthermore, the CARS algorithm interprets most of the effective information by the weights, and it can find sensitive variables and highlight core feature in the whole wavelength optimization. Itprovides a foundation for the selection of reliable features in the SR. However, the validity of reconstruction from a data perspective needs to be further discussed.

3.5. Multivariate correlation analysis

In fact, RGB image contains wide-band information in the visible range of 400–780 nm, with the band range of R, G, B channel are 620–780 nm, 500–560 nm and 430–480 nm, respectively. It should be emphasized that the feature wavelength selected in HSI has a distribution range of 780–1000 nm. Therefore, it is important to study whether the SR techniques can accurately reconstruct the spectral in the NIR range. In theory, highly correlated data can be used to establish relationships between them using machine learning methods. The feasibility of the SR algorithm in the Vis-NIR range was explored in this study. The correlation between three-components of RGB and HSI sub-bands (430–480 nm, 500–560 nm, 620–780 nm, 780–1000 nm and 400–1000 nm) was calculated and evaluated.

Fig. 4 shows the correlation between RGB components and HSI sub-bands, and the multivariate correlation analysis is used to add credibility to the reconstruction research. SPSS was used to test and analyze the correlation between the parameters, and a multivariate correlation scatter matrix was formed by combining the content histogram, scatter plot and Pearson correlation analysis plot. The analysis of the content histogram showed that the average spectral of the producing region from Jingyuan was higher than that of other producing regions, which was consistent with the spectral reflectance index visualization. Meanwhile, the scatter plot showed that there were more outliers in the

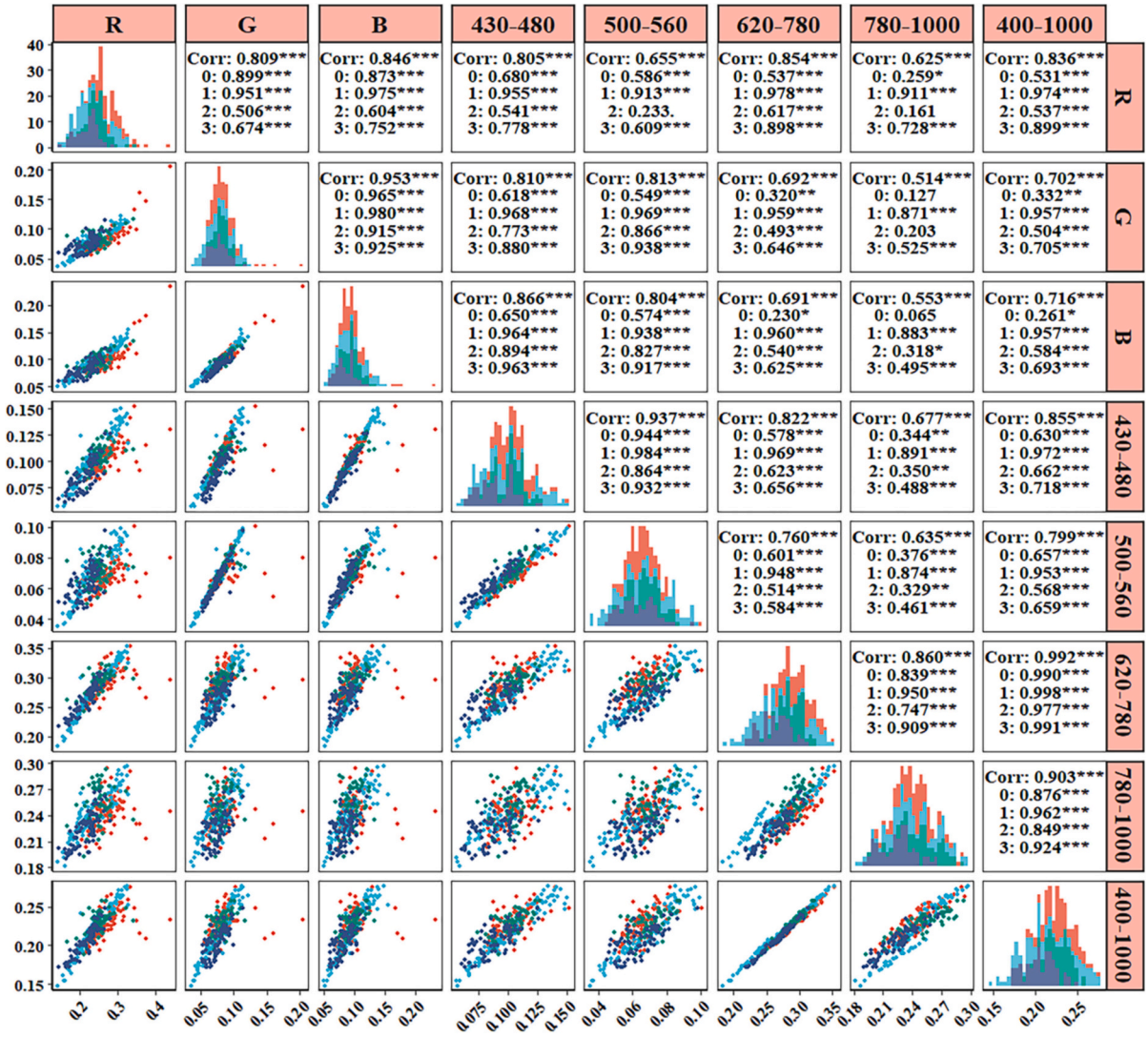


Fig. 4. The diagrams of multivariate correlation scatter matrix. Note: In the histogram and scatter plot of the content, red represents the producing region from Jingyuan, cyan represents the producing region from Longde, green represents producing region from Xiji, and brown represents the producing region from Haiyuan. * $P < 0.05$, ** $P < 0.01$, *** $P < 0.001$. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

spectral of the producing region from Jingyuan, which leaded to a decrease in the correlation between RGB and other components. In addition, by analyzing Pearson correlation, the three-components of RGB had the highest correlation with the spectral of the corresponding range. It was obviously observed that the correlation between 780 and 1000 nm and all the variables was the lowest, but the value of correlation coefficient was above 0.5, which could be considered as a moderate correlation between the two variables. This provided practical theoretical support for reconstructing RGB-to-HSI in the 400–1000 nm range.

3.6. Spectral reconstruction

3.6.1. Training details of spectral reconstruction model

Fig. 5(a) shows the convergence of MRAE values and LOSS curves for 11 models in the training set. In the LOSS curves, All models converged rapidly after 200 epochs. The convergence effect of MST, MST++ and MPRNet models was the best, and the average loss was 0.13. It should be noted that there were significant variations in the loss values of different algorithms for MRAE. Among, the MIRNet algorithm had the best loss value, about 0.1002, while the HINet algorithm had the worst performance, with an evaluation loss of about 0.3962.

3.6.2. Quality evaluation of spectral reconstruction

Based on the minimum LOSS and MRAE values of 11 models in the data set, the corresponding RMSE, PSNR and SSIM values are determined as shown in Table 1. All SOTA models showed good reconstruction performance. Among, in terms of RMSE and PSNR, MST++ showed the best performance, with RMSE and PSNR values of 0.015 and 36.70, respectively. Compared with HSCNN++ (*NTIRE 2018* winner) and AWAN (*NTIRE 2020* winner), the value of PSNR for MST++ was improved by 6.05 and 3.40 dB (dB), respectively. This result was better than the SOTA method used by Cai, Lin, Lin, et al. (2022) on the *NTIRE 2022* HSI data set (MST++ testing set PSNR is 34.32). It showed that MST++ algorithm could capture more spatial features. In previous investigations found that the SR algorithm for different data sets had different, and the performance displayed also varies. For instance, Fu et al. (2022) used the HSCNN++ algorithm achieved the best performance in corn disease detection, but showed an opposite trend to the reconstructed results for this data set. In addition, Zamir et al. (2021) and Hu et al. (2023) conducted SR for marine environment and open data set, respectively, and the results showed that the reconstruction performance had specific sample adaptability. The result showed that extended HSI data was important to generalize SR models.

To intuitively show the reconstruction performance advantages

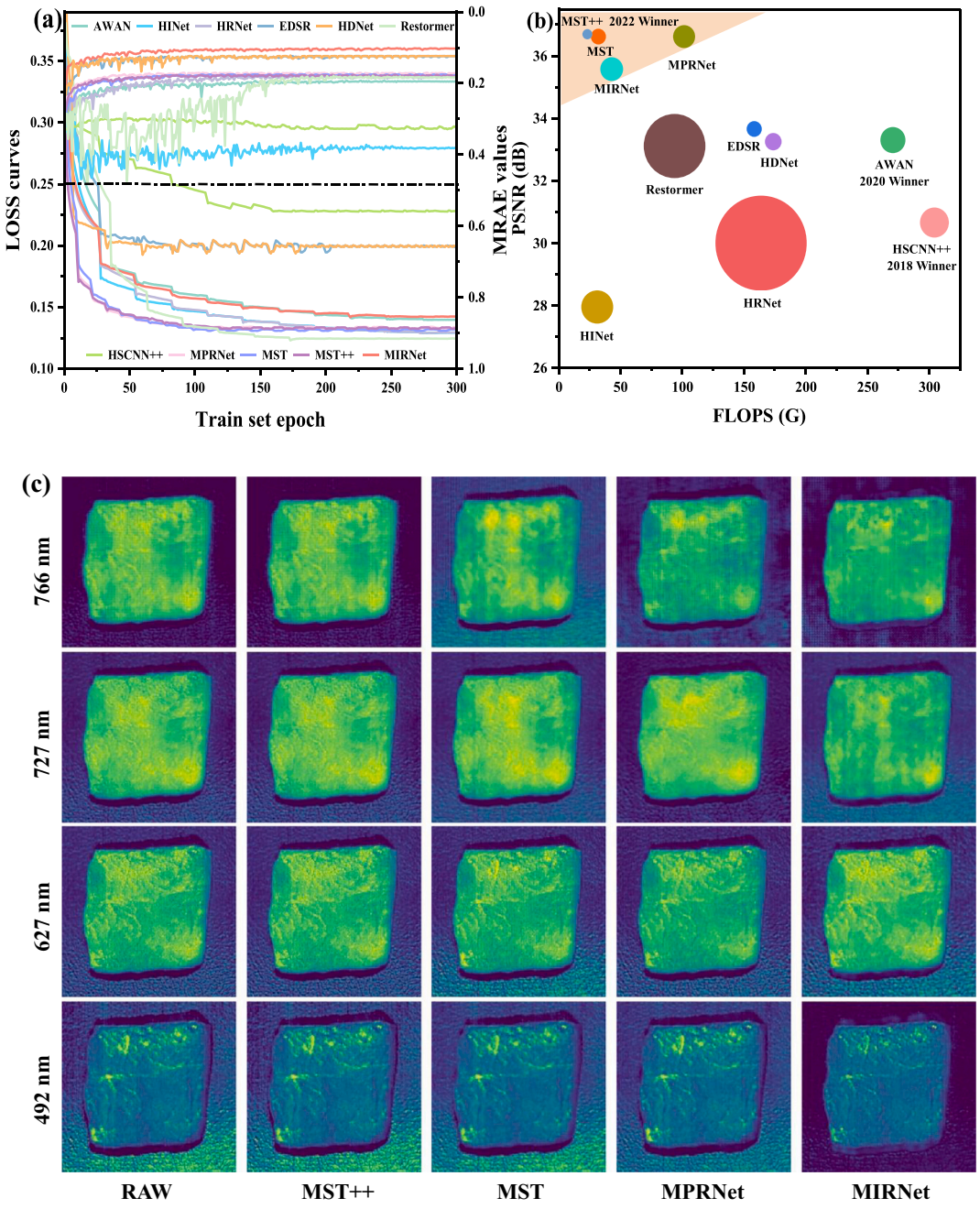


Fig. 5. Quality evaluation for spectral reconstruction. (a) LOSS curve and MRAE value of the training set spectral reconstruction algorithm; (b) comparisons of PSNR-Params-FLOPS for all spectral reconstruction algorithms; (c) reconstruction images of four optimal models under selected frequency bands.

Table 1
Comparison of multivariate evaluation indexes for 11 spectral reconstruction algorithms.

Model	Params	FLOPS	Train set				Test set			
			MRAE	RMSE	PSNR	SSIM	MRAE	RMSE	PSNR	SSIM
HINet	5.21	31.04	0.3962	0.0452	27.942	0.8037	0.3265	0.0553	27.093	0.7991
HRNet	31.70	163.81	0.1727	0.0345	30.001	0.8297	0.1955	0.0407	30.028	0.8298
HSCNN+	4.65	304.45	0.3086	0.0312	30.653	0.8315	0.2083	0.0318	30.703	0.8355
Restormer	15.11	93.77	0.2392	0.0229	33.114	0.8421	0.1368	0.0254	32.505	0.8411
HDNet	2.66	173.81	0.1309	0.0226	33.257	0.8452	0.1309	0.0229	33.257	0.8458
AWAN	4.04	270.61	0.2068	0.0207	33.303	0.8482	0.1291	0.0244	33.389	0.8577
EDSR	2.42	158.32	0.1220	0.0212	33.654	0.8608	0.1237	0.0223	33.479	0.8601
MIRNet	3.75	42.95	0.1002	0.0172	35.574	0.9059	0.1002	0.0176	35.527	0.9047
MPRNet	3.62	101.59	0.1733	0.0150	36.617	0.9220	0.0977	0.0169	35.910	0.9135
MST	2.45	32.07	0.1797	0.0149	36.616	0.9220	0.0995	0.0156	36.870	0.9201
MST++	1.62	23.05	0.1780	0.0149	36.700	0.9253	0.0972	0.0156	36.966	0.9251

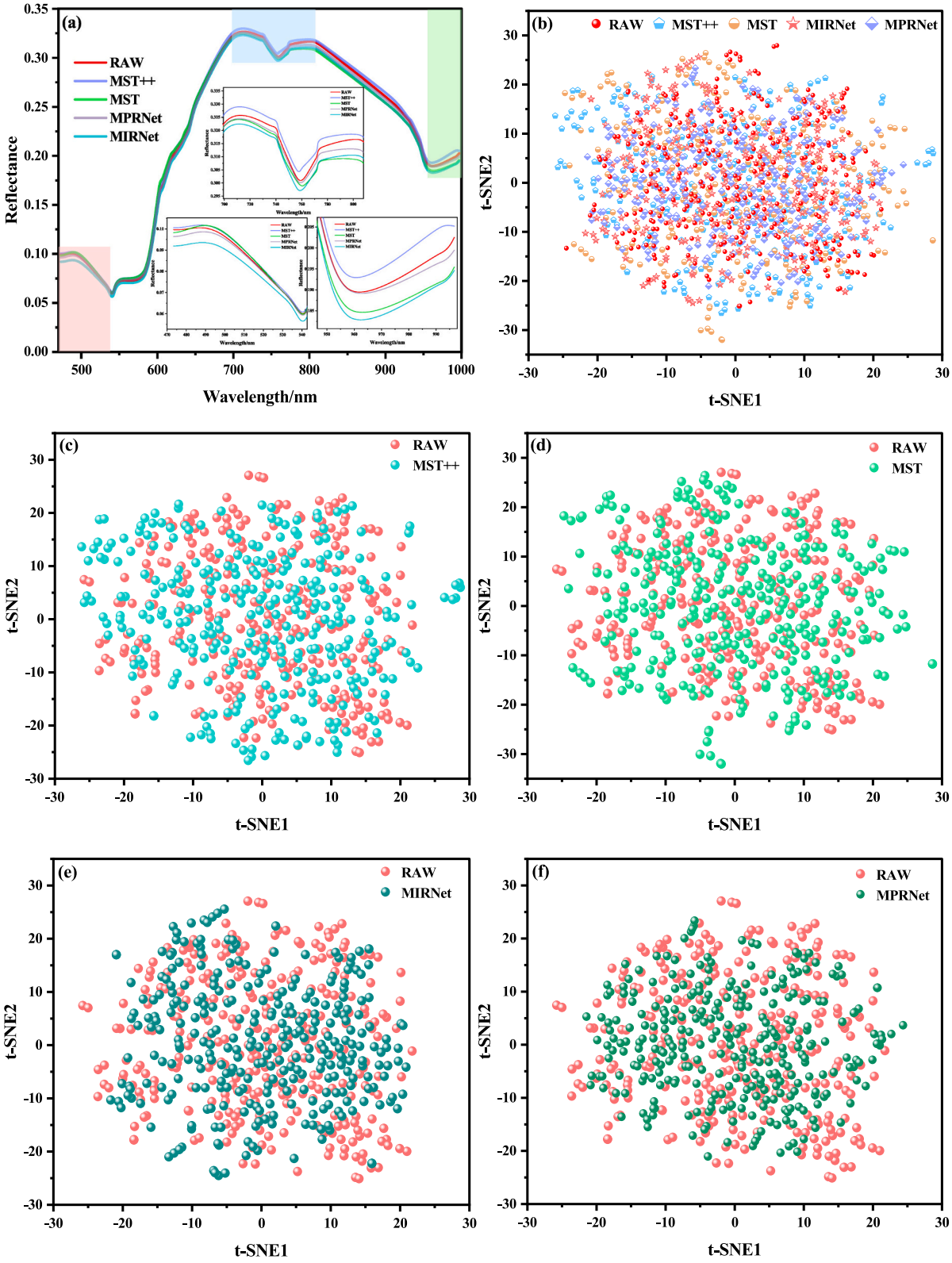


Fig. 6. Consistency analysis for spectral reconstruction images. (a) The average spectral reflectance curve of the optimal reconstructed model; (b) visualization of optimal reconstruction model by t-SNE dimension reduction; (c) consistent distribution of RAW and MST++ data; (d) consistent distribution of RAW and MST data; (e) consistent distribution of RAW and MIRNet data; (f) consistent distribution of RAW and MPRNet data.

Table 2
Comparison of Raw-HSI, RGB and SR-HSI modeling performance based on lightweight Transformer.

Index	Calibration set (n = 216)		Cross validation set (n = 72)		Prediction set (n = 72)		RPD
	R_C^2	RMSEC	R_{cv}^2	RMSECV	R_P^2	RMSEP	
Raw-HSI	0.8710	0.1504	0.8569	0.1585	0.8526	0.1590	2.69
RGB	0.6891	0.4538	0.6435	0.4561	0.6007	0.4631	0.92
MST++	0.8658	0.1624	0.8444	0.1746	0.8422	0.1752	2.46
MST	0.8572	0.1561	0.8389	0.1923	0.8412	0.1785	2.40
MIRNet	0.8441	0.1748	0.8340	0.1965	0.8218	0.1984	2.21
MPRNet	0.8543	0.1572	0.8186	0.1993	0.8246	0.1975	2.30

between models, the quality attributes (computational cost, reconstruction quality, and performance efficiency) were compared comprehensively, and the best model was selected for follow-up analysis. The comparison of PSNR-Params-FLOPS for different algorithms is provided in Fig. 5(b). The vertical axis is PSNR (performance), the horizontal axis is FLOPS (computational cost), and the circle radius is Params (memory cost). It can be seen that MST++, MST, MIRNet, and MPRNet were distributed in the top-left corner, exhibiting the extreme efficiency advantages of the methods.

The four optimal models based on the above comprehensive quality attributes had practical significance for further evaluating the perceived quality of beef SR. As shown in Fig. 5(c), reconstructed images of the MST++, MST, MIRNet and MPRNet models are generated at four selected bands (492, 627, 727, and 766 nm). The results showed that the overall reconstructed images performed well in terms of visual authenticity and detail preservation. Especially, the reconstructed images at 627, 727, and 766 nm had smaller errors and better agreement with the authentic spectral images. Because the RGB images not only contain the information of red, green and blue channels, but also cover part of the spectral information in the visible range. In addition, in the texture details of the image, MST++ and MST are closest to the RAW image, while MIRNet differs the most, especially at 766 nm. In summary, the validity of spectral reconstruction quality had been fully verified, but the study of SR image consistency distribution needed to be further explored.

3.6.3. Consistency analysis for spectral reconstruction images

To assess the consistency between the original and reconstructed images, the sliding window method was used to calculate SSIM. After traversing the entire image, take the average of all window values as the SSIM value of the image (Table 1). The results showed that the SSIM values of all reconstructed models were greater than 0.8, demonstrating the reliability of reconstructed image from the data perspective. The SSIM values of MST++, MST, MIRNet, and MPRNet algorithms were all greater than 0.9, indicating that more effective spatial features were preserved. This was consistent with the conclusion that MST++, MST, MIRNet, and MPRNet had the best comprehensive properties in SR quality evaluation, which provided support for the rationality of RGB-to-HSI data.

To further show the reconstruction details, the average spectral curves of 31 spectral channels of the original and reconstructed HSI curve of the four reconstruction algorithms were compared (Fig. 6(a)). It can be observed that the reconstruction error between 440–540 nm and 930–997 nm was the most obvious. This was mainly the influence of instrument drift and noise at both ends of the collected spectral. Similar phenomena were observed in the study by Fu et al. (2022) and Hu et al. (2023). It was worth noting that SR was best at 550–700 nm. Furthermore, the average spectral curve of the MST++ algorithm at 700–780 nm was significantly higher than the original. This was because that MST++ performed better in edge clarity and maintaining spatial smoothness in uniform areas (Cai, Lin, Lin, et al., 2022). In addition, the reconstruction fit in the NIR range (750–1000 nm) was lower than in the Vis range (400–750 nm). This was because the RGB image did not contain spectral information above 780 nm, but there was a certain correlation between the bands, which was consistent with the conclusion of multivariate correlation analysis.

Finally, to visually show the difference between the actual and reconstructed spectral data, the t-SNE was used for data reduction and visualization (Fig. 6(b)). The results showed that the spatial distribution of reconstructed spectral data was very close to the actual spectral data, and the two were in an inseparable state. It indicated that the reconstructed spectral data had highly similar characteristics to the real spectral data. In addition, it is observed from Fig. 6(c-f) that the reconstructed data points for MST++ and MST are decentralized, while the data points for MIRNet and MPRNet are centralized. It was showed that the MST series model can reduce the difference between the reconstructed HSI and the actual data by adding richer context information on the basis of ensuring local detail integrity. This is mainly due to the outstanding performance of MST series models in the modeling of spectral self-similarity and dependence. However, the relationship between the reconstructed spectral data and the corresponding physicochemical properties needed to further validation and analysis are required.

3.7. Analysis for spectral quantitative model

The comprehensive reconstruction quality evaluation showed that MST++, MST, MIRNet, and MPRNet algorithms performed the best in beef spectral reconstruction. Quantitative results of Raw-HSI, RGB and SR-HSI images based on Lightweight Transformer are shown in Table 2. The results showed that the Raw-HSI image achieved the best prediction performance with R_P^2 of 0.8526 and RPD of 2.69. This showed that the real spectral image had more accurate and comprehensive scene characteristics. In comparison, the prediction results of RGB images needed to be further improved (R_P^2 = 0.6007 and RPD = 0.92). This was consistent with the conclusion of Arad et al. (2018) that RGB images could be brighter or dimmer was depended on the camera exposure settings and the varying illumination intensity of the scene, which corresponded to a linearly scaled truth-value spectrum. This meant a linearly scaled HSI and its RGB counterpart was also a physically valid ground-truth pair (Arad et al., 2018). It should be emphasized that the Lightweight Transformer models of the four SR-HSI images showed good predictive performance. Among, the reconstruction effects of MST and MST++ of the Transformer-based model was R_P^2 = 0.8422 and R_P^2 = 0.8412, respectively, which were closest to the Raw-HSI result. Itwas because that the combination of MST and MST++ with a Multi-stage learning scheme and Spectral-wise Multi-head Self-attention could accurately obtain the long-distance feature correlation between all pixels (Cai, Lin, Lin, et al., 2022). The coefficient of determination results of Natural image restoration model based on MIRNet and MPRNet reconstruction resultsin prediction set reaches above 0.82. This showed that the Lightweight Transformer model could obtain rich feature details and spectral information in the reconstructed image. Compared with the four reconstruction models, the Lightweight Transformer model had potential advantages in extracting the characteristic spectral information of MST series reconstruction images. This was related to the fact that the global information extracted by branches of the Transformer-based framework was fed into the Lightweight Transformer to

enhance global awareness and maximize the retention of feature information.

4. Conclusion

The feasibility of low-cost application of HSI through RGB image reconstruction in complex food inspection system was discussed in this study. Taking the detection of Glu index in beef as an example, the feasibility of reconstruction were evaluated by 11 SOTA algorithms combined with multivariate analysis. Firstly, the compressive sensing characteristics and differences between channels were visually comparied by utilizing the spectral reflectance index. A otal of 31 bands of enhanced feature information were extracted by CARS algorithm. Then, multivariate correlation scatter matrix was adopted to provide RGB-to-HSI theoretical support from the data perspective. In addition, comprehensive quality attribute (PSNR-Params-FLOPS) was proposed to determine the optimal reconstruction model. Furthermore, SSIM value and t-SNE consistency were introduced to evaluate reconstruction quality. The results showed that all SOTA models showed good reconstruction performance. The algorithms of MST++, MST, MIRNet, and MPRNet were selected to better adapt to spectral reconstruction of beef based on comprehensive quality attributes. Among, MST++ showed the best performance, with the RMSE, PSNR, and SSIM values of 0.015, 36.70, and 0.9253, respectively. Finally, the quantitative analysis of four comprehensive optimal reconstruction models was established based on the Lightweight Transformer model. Compared with the Raw-HSI (R_P^2 = 0.8526), the reconstructed MST++ model had the best performance, and the R_P^2 was e 0.8422. This study provides a new perspective for HSI quantitative analysis.

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Ethics statement

This article does not contain any studies with human or animal subjects.

CRediT authorship contribution statement

Fujia Dong: Writing – original draft, Methodology, Data curation. **Ying Xu:** Validation, Methodology. **Yingkun Shi:** Investigation. **Yingjie Feng:** Validation. **Zhaoyang Ma:** Software, Investigation. **Hui Li:** Supervision. **Zhongxiong Zhang:** Software. **Guangxian Wang:** Investigation. **Yue Chen:** Project administration. **Jinhua Xian:** Investigation. **Shichang Wang:** Investigation. **Songlei Wang:** Project administration, Methodology, Investigation. **Weiguo Yi:** Formal analysis, Conceptualization.

Declaration of competing interest

We declare that none of the work contained in this manuscript is published in any language or currently under consideration at any other journal, and there are no conflicts of interest to declare. All authors have contributed to, read, and approved this submitted manuscript in its current form.

Data availability

Data will be made available on request.

Acknowledgments

Thank you for everything my grandfather has done. This article commemorates the one-year anniversary of his passing.

Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.foodchem.2024.141543.

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